

Author Contribution Statement

Conservativeness of the CUEP Energy Bound in IBR-Penetrated Networks: Proof, Monotonicity, and Application to the SAMBPS Protection Framework

CRedit Taxonomy

Contribution	A. V. Eluvathingal	R. K. Prasad
Conceptualisation	Lead	Supporting
Formal analysis	Lead	Supporting
Funding acquisition	—	Lead
Investigation	Lead	—
Methodology	Lead	Supporting
Project administration	—	Lead
Software (simulation)	Lead	—
Supervision	—	Lead
Validation	Lead	Supporting
Visualisation	Lead	—
Writing — original draft	Lead	—
Writing — review & editing	Supporting	Lead

Plain-Language Statement (for portal upload)

Arun V. Eluvathingal: Identified the gap in the CUEP conservativeness proof for IBR-penetrated systems; formulated the IBR-modified energy function; proved Theorem 1 (conservativeness), Lemmas 1–2 (CUEP angle and energy monotonicity), Theorem 2 (monotone conservatism gap), and Proposition 1 (SAMBPS C13 zero-false-trip result); designed and executed the IEEE 39-bus simulation study (1500 scenarios); produced all figures and tables; wrote the manuscript draft.

R. K. Prasad: Supervised the research; provided guidance on the CUEP method literature and the appropriate assumptions for IBR current-limiter modelling; reviewed and edited the final manuscript; secured funding through the SAMBP project.

Conflict of interest: The authors declare no conflict of interest.

Code availability: MATLAB/Simulink scripts for the IEEE 39-bus test system with IBR injection, CUEP location, and energy-function computation will be deposited in the IIT Madras institutional repository upon acceptance.